

DNA computing

Tags:

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Codes

AI Code Tips

Description

DNA computing is a form of computing which uses DNA, biochemistry and molecular biology, instead of the traditional silicon-based computer technologies. DNA computing, or, more generally, molecular computing, is a fast developing interdisciplinary area. Research and development in this area concerns theory, experiments and applications of DNA computing. DNA computing is fundamentally similar to parallel computing in that it takes advantage of the many different molecules of DNA to try many different possibilities at once. So, many NP problem can be solved through DNA computing, such as Hamiltonian path problem, SAT problems, and so on.

The core reaction of DNA computing is the specific hybridization, which may results in incorrect or undesirable computations. Therefore, so far much works have focused on designing the DNA sequences to make the molecular computation more reliable.

The DNA encoding problem can be described as follows: the encoding alphabet of DNA sequence is the set of nucleic acid bases $\Sigma=\{A, C, G, T\}$, and in the encoding set Z of DNA strands whose length is L , $Z=\Sigma^L=\{b_1b_2...b_L \mid b_i \in \Sigma, i=1,2,...,L\}$, and $|Z|=4^L$, search the subset W of Z which satisfies (any subset of Z can be treat as a code set):

(1) $\forall x_i \in W$, the total number of A、C in x_i is $L/2$;

(2) $\forall x_i, x_j \in W$, the hamming distance (the **Hamming distance** between two strings of equal length is the number of positions for which the corresponding symbols are different) between x_i and x_j is at least D.

In order to simplify the encoding problem, we don't want you to find such a subset with the biggest size, we only want you to find the best subset satisfying the two conditions. The strategy to compare two subset is as follows:

Suppose W_1, W_2 are two subset satisfying the two conditions. If $W_i (i=1, 2)$ is a subset of $W_j (j=2, 1)$, then $W_j (j=2, 1)$ is better than $W_i (i=1, 2)$. Otherwise, we firstly order the elements of them independently based on lexicographic. then, we get the first element x_1 from W_1 and the first element y_1 from W_2 . We consider W_1 to be better than W_2 if x_1 is in front of y_1 in lexicographic. on the contrary, we think W_2 is better than W_1 if x_1 is behind y_1 . On the other hand, if x_1 is the same as y_1 , we then get the second element x_2 from W_1 and the second element y_2 from W_2 , if they are also the same, we will get the next pairs until we can evaluate them.

For examples :

(1) $W_1 = \{A, C, G, T\}, W_2 = \{A, T\}$

W_1 is better than W_2 because W_2 is a subset of W_1 .

(2) $W_1 = \{AC, CT, AT\}, W_2 = \{AG, AC\}$

Firstly, we order them, so $W_1 = \{AC, AT, CT\}, W_2 = \{AC, AG\}$

Then W_2 is better than W_1 because AG is in front of AT.

Now, we will give you the length L and the distance D, can you find the best code set.

Input

There are several cases. Each case contains two integers, L ($0 < L \leq 25$) and D ($0 \leq D \leq L$).

Output

For each case, output several lines, the first line is the size of the code set, then print all the elements of the code set in lexicographic. A element per line.

Samples

input	Copy
2 2	
output	Copy
4 AG CT GA TC	

Source

3rd Central South China Programming Contest

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Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

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